

1. Identification of The Substance or Mixture and The Supplier

Product	
Substance or Trade Name	Hexane
Recommended Use	Reaction diluent, solvent, and chemical industrial
Details of the Supplier of the Safety Data Sheet	
Company Name	Bangchak Corporation Public Company Limited
Address	2098 M Tower Building, 8th floor, Sukhumvit Road, Phra Khanong Tai, Phra Khanong, Bangkok 10260 Thailand
Telephone	+66 2335 4999
Fax	+66 2016 3991
Emergency Number	+66 2335 8888

2. Hazards Identification

Classification of the Mixture according to Globally Harmonized System (GHS) Standards.

Flammable Liquid	Category 2
Skin Corrosion/Irritation	Category 2
Eye Irritation	Category 2B
Reproductive Toxicity	Category 2
Aspiration Hazard	Category 1
Specific Target Organ Toxicity, Single Exposure	Category 3 (Central nervous system)
Specific Target Organ Toxicity, Repeated Exposure	Category 2 (Central nervous system)

GHS Label Elements



Signal Word: Danger

Hazard Statements:

- H225 – Highly flammable liquid and vapor.
- H304 – May be fatal if swallowed and enters airways.
- H315 – Causes skin irritation.
- H336 – May cause drowsiness or dizziness.
- H361 – Suspected of damaging fertility or the unborn child.
- H373 – May cause damage to organs through prolonged or repeated exposure.

Precautionary Statements:

- P201 – Obtain special instructions before use.
- P202 – Do not handle until all safety precautions have been read and understood.
- P210 – Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
- P233 – Keep container tightly closed.
- P240 – Ground and bond container and receiving equipment.
- P241 – Use explosion-proof electrical/ ventilating/ lighting equipment.
- P242 – Use non-sparking tools.
- P243 – Take action to prevent static discharges.
- P260 – Do not breathe dust/ fume/ gas/ mist/ vapors/ spray.
- P264 – Wash skin thoroughly after handling.
- P271 – Use only outdoors or in a well-ventilated area.
- P280 – Wear protective gloves/ protective clothing/ eye protection/ face protection.
- P301 + P310 – IF SWALLOWED: Immediately call a poison center/ doctor.
- P303 + P361 + P353 – IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water.
- P304 + P340 + P312 – IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a poison center/ doctor if you feel unwell.
- P308 + P313 – IF exposed or concerned: Get medical advice/ attention.
- P331 – Do NOT induce vomiting.
- P332 + P313 – IF skin irritation occurs: Get medical advice/ attention.
- P362 + P364 – Take off contaminated clothing and wash it before reuse.
- P370 + P378 – In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.
- P403 + P233 – Store in a well-ventilated place. Keep container tightly closed.
- P403 + P235 – Store in a well-ventilated place. Keep cool.
- P405 – Store locked up.
- P501 – Dispose of contents/ container to an approved waste disposal plant.

Other hazards

The following percentage of the mixture consists of ingredient(s) with unknown acute toxicity: 59.9999 %

3. Composition/Information on Ingredients

This material is defined as a mixture.

Hazardous Substance(s) or Complex Substance(s) required for disclosure

No.	Component	CAS No.	Content (%)
1	Naphtha (Petroleum), Hydrotreated Light	64742-49-0	100

Hazardous Constituent(s) Contained in Complex Substance(s) required for disclosure

No.	Component	CAS No.	Content (%)
1	Cyclohexane	110-82-7	≤ 3
2	Hexane (Mixture or Isomers)	Excl. n-Hexane	30 - 45
3	Methylcyclopentane	96-37-7	10 - 20
4	n-Hexane	110-54-3	40 - 60

4. First Aid Measures

Inhalation	Immediately remove from further exposure. Get immediate medical assistance. For those providing assistance, avoid exposure to yourself or others. Use adequate respiratory protection. Give supplemental oxygen, if available. If breathing has stopped, assist ventilation with a mechanical device.
Skin Contact	Wash contact areas with soap and water. Remove contaminated clothing. Launder contaminated clothing before reuse.
Eye Contact	Flush thoroughly with water. If irritation occurs, get medical assistance.
Ingestion	Seek immediate medical attention. Do not induce vomiting.
Note to Physician	If ingested, material may be aspirated into the lungs and cause chemical pneumonitis. Treat appropriately. This material, or a component, may be associated with cardiac sensitization following very high exposures (well above occupational exposure limits) or with concurrent exposure to high stress levels or heart-stimulating substances like epinephrine. Administration of such substances should be avoided.

5. Fire Fighting Measures

Suitable Extinguishing Media:	Foam, dry chemical, or carbon dioxide (CO ₂) to extinguish flames.
Unsuitable Extinguishing Media:	Straight streams of water.

Required Special Protective Equipment for Fire-fighters:	Evacuate area. If a leak or spill has not ignited, use water spray to disperse the vapors and to protect personnel attempting to stop a leak. Prevent run-off from fire control or dilution from entering streams, sewers or drinking water supply. Fire-fighters should use standard protective equipment and in enclosed spaces, self-contained breathing apparatus (SCBA). Use water spray to cool fire exposed surfaces and to protect personnel.
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6. Accidental Release Measure

Personal Precautions:	Use personal protective equipment. Ensure adequate ventilation. Remove all sources of ignition. Evacuate personnel to safe areas. Beware of vapors accumulating to form explosive concentrations. Vapors can accumulate in low areas.
Environmental Precautions:	Prevent products from entering drains. Prevent further leakage or spillage if safe to do so. If the product contaminates rivers and lakes or drains inform respective authorities.
Methods for Cleaning:	Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).

7. Handling and Storage

Handling:	Avoid all personal contact. Prevent exposure to ignition sources, for example use non-sparking tools and explosion-proof equipment. Potentially toxic/irritating fumes/vapor may evolve from heated or agitated material. Use only with adequate ventilation. Prevent small spills and leakage to avoid slip hazard. Material can accumulate static charges which may cause an electrical spark (ignition source). Use proper bonding and/or ground procedures. However, bonding and grounds may not eliminate the hazard from static accumulation. Consult local applicable standards for guidance. Additional references include American Petroleum Institute 2003 (Protection Against Ignitions Arising out of Static, Lightning and Stray Currents) or National Fire Protection Agency 77 (Recommended Practice on Static Electricity) or CENELEC CLC/TR 50404 (Electrostatics - Code of practice for the avoidance of hazards due to static electricity).
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Storage:	Ample fire water supply should be available. A fixed sprinkler/deluge system is recommended. The type of container used to store the material may affect static accumulation and dissipation. Keep container closed. Handle containers with care. Open slowly in order to control possible pressure release. Store in a cool, well-ventilated area. Outside or detached storage preferred. Storage containers should be grounded and bonded. Fixed storage containers, transfer containers and associated equipment should be grounded and bonded to prevent accumulation of static charge.
Storage Temperature:	Ambient temperature
Other Information:	No data available.

8. Exposure Controls/Personal Protection

Engineering Measures:	The level of protection and types of controls necessary will vary depending upon potential exposure conditions. Controlled measures to consider: Adequate ventilation should be provided so that exposure limits are not exceeded. Use explosion-proof ventilation equipment.
Personal Protection	
Respiratory Protection:	If engineering controls do not maintain airborne contaminant concentrations at a level which is adequate to protect worker health, an approved respirator may be appropriate. Respirator selection, use, and maintenance must be in accordance with regulatory requirements, if applicable. Types of respirators to be considered for this material include: Half-face filter respirator Type A filter material.
Eye Protection:	If contact is likely, safety glasses with side shields are recommended.
Hand Protection:	Any specific glove information provided is based on published literature and glove manufacturer data. Glove suitability and breakthrough time will differ depending on the specific use conditions. Contact the glove manufacturer for specific advice on glove selection and breakthrough times for your use conditions. Inspect and replace worn or damaged gloves. The types of gloves to be considered for this material include: Chemical resistant gloves are recommended: Nitrile.
Hygiene Measures:	Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants. Discard contaminated clothing and footwear that cannot be cleaned. Practice good housekeeping.

9. Physical and Chemical Properties

Physical:	Liquid
Color:	Colorless
Odor:	Slight
Flammability:	Not available
Density:	0.68 kg/L at 15.0 °C
Pour Point:	-108 °C
Freezing Point:	-95 °C
Flash Point:	-28 °C
Boiling Point:	65 to 68 °C
Melting Point:	No data available.
Auto-ignition Temperature:	267 °C (Extrapolation)
Decomposition Temperature:	No data available.
Vapor Pressure:	18 kPa at 20.0 °C
Flammable Limits:	Min (LEL) % : 1.2 % v/v Max (UEL) % : 8.0 % v/v
Evaporation Rate:	14 (n-Butyl acetate = 1)
pH:	No data available.
Solubility:	Negligible in water
Viscosity:	0.4 cSt at 40.0 °C
Molecular Weight:	86 g/mol (Calculation)
Coefficient of Thermal Expansion:	0.00135 per 1 °C (Calculation)
Octanol-water Partition Coefficient (Log P_{ow}):	> 4 (Estimation)

10. Stability and Reactivity

Stability:	Material is chemically stable under standard ambient conditions (room temperature).
Materials to Avoid:	Strong oxidizers, rubber, various plastics.
Hazardous Decomposition Products:	In the event of fire: see section 5.

11. Toxicological Information

Acute Toxicity	
Oral	LD50 (Rat, Male and Female): 16.000 mg/kg (OECD Test Guideline 401)

Dermal	LD50 (Rabbit, Male): > 2.000 mg/kg (OECD Test Guideline 402)
Inhalation	LC50 (Rat): 4 h - 172 mg/L (RTECS)
Skin Corrosion/Irritation	Rabbit: 24 h (OECD Test Guideline 404)
Eye Damage/Irritation	Rabbit: No irritation - 72 h (OECD Test Guideline 405)
Germ Cell Mutagenicity	No data available
Carcinogenicity	No data available
Reproductive Toxicity	Overexposure may cause reproductive disorder(s) based on tests with laboratory animals. Suspected human reproductive toxicant Suspected of damaging fertility. Suspected of damaging fertility.
Specific Target Organ Toxicity - Single Exposure	May cause drowsiness or dizziness - Central nervous system.
Specific Target Organ Toxicity - Repeated Exposure	May cause damage to organs through prolonged or repeated exposure - Nervous system.
Aspiration hazard	LC50 (Mice, 4 hrs.); 48000 ppm by inhalation. LD50 (Rats); Orally: 32.0 g/kg (Couri, Milks)

12. Ecological Information

Ecotoxicity	Expected to be toxic to aquatic organisms. May cause long-term adverse effects in the aquatic environment.
Persistence and Degradability	Expected to be readily biodegradable.
Bio-accumulative Potential	No data available
Mobility	Highly volatile, will partition rapidly to air. Not expected to partition to sediment and wastewater solids.
Other Adverse Effects	An environmental hazard cannot be excluded in the event of unprofessional handling or disposal. Toxic to aquatic life with long lasting effects.

13. Disposal Considerations

Waste Disposal	Dissolve or mix the material with a combustible solvent and burn in a chemical incinerator equipped with an afterburner and scrubber. All federal, state, and local environmental regulations must be observed.
Container Disposal	Empty remaining contents. Dispose of as unused product. Do not re-use empty containers. Do not burn, or use a cutting torch on, the empty drum.

14. Transport Information

Dangerous for Conveyance unless the Flash Point is known to be greater than 61 °C.

Transportation of Dangerous Goods (TDG)

UN Number	UN 1208
Proper Shipping Name	Hexanes
Hazard Class	3
Packing Group	II

International Air Transport Association (IATA)

UN Number	UN 1208
Proper Shipping Name	Hexanes
Hazard Class	3
Packing Group	II
Environmental Hazard	No

International Maritime Dangerous Goods (IMDG)

UN Number	UN 1208
Proper Shipping Name	Hexanes
Hazard Class	3
Packing Group	II
Marine Pollutant	Yes

15. Regulatory Information

This material is considered hazardous according to the classification criteria of the Hazard Classification and Communication System for Hazardous Materials BE 2555 but not regulated by Hazardous Substance Act BE 2535.

16. Other Information

Issue Date: 27 May 2009
Revision Date: 24 March 2026

Reference

1. National Institute of Technology and Evaluation (SAFE NITE)
http://www.safe.nite.go.jp/english/ghs/ghs_index.html
2. Globally Harmonized System of Classification and Labelling of Chemical (GHS), United Nation, 2011

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